

Sled Vibration Monitoring Guide

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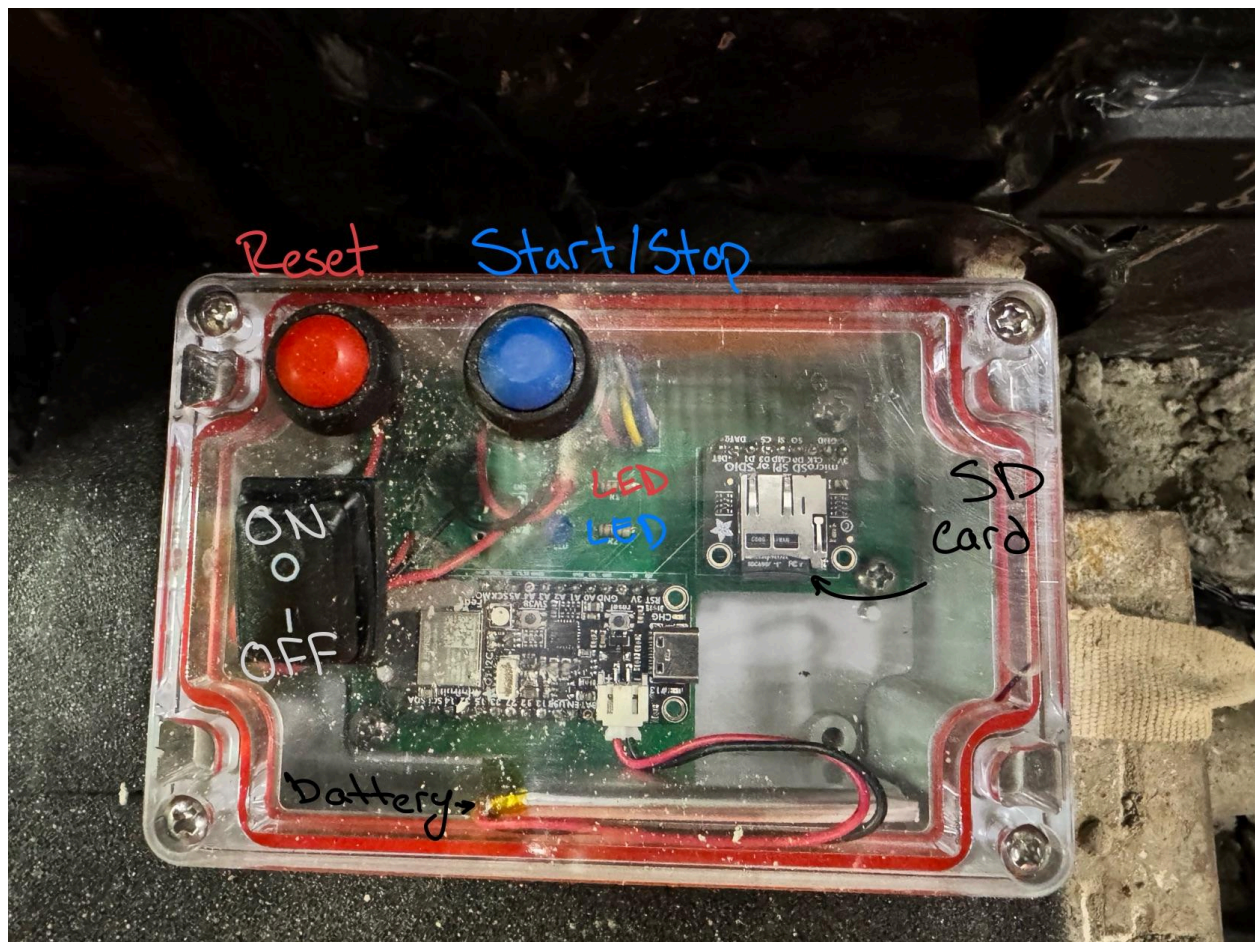
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Labeled Diagram



Power switch - turns device on/off

Red button - resets the device, useful for making sure SD card works

Blue button - starts and stops recording

Blue LED

Solid - it is in the ready state, waiting for the blue button press.

Flashing - there is an SD card issue, try resetting

Red LED

Solid - it is in the recording state, press blue button to finish recording

How to operate the device

1. Flip the power switch to ON (a blue light will come on)
2. If the blue light is flashing, it indicates a problem with the SD card, either because it is not inserted or has not been initialized properly. Press the red button to reset the system.
3. While the blue light is solid, the device is in the ready state. Press the blue button to start a recording.
4. Press the blue button again to stop the recording. (Both LEDs will flash)
5. The device will then return to the ready state (blue light solid)

There is about 5 seconds after pushing the blue button to start recording during which pushing it again will do nothing. After 5 seconds, pushing the button will stop the recording as normal.

How to view live data

1. While the device is powered on, join the wifi network *ESP32-Graph* with the password *12345678*
2. Using a browser, type into the search bar *192.168.4.1* and search
3. If it asks you to use cellular data, allow it
4. The graphs should then begin to populate as the sensor runs

There is a chance that it won't work immediately, so refresh occasionally and ensure you are connected to the network until the graphs load.

How to access logging files

1. Unscrew the 4 Phillips head screws on the corner of the brain box

2. Gently flip up the lid toward the side where the wires exit the box (it won't come all the way off since there are wires connecting the buttons)
3. Push the SD card in, you will hear a click, and the SD card will release
4. Remove the SD card from the box
5. Plug the SD card into a computer
6. Files are labeled in sequential order as such: *quakeData_13.csv*. Do not delete or move the *runcount.txt* file
7. Copy or move your files as needed
8. Return the SD card to the brain box, ensuring you push it in enough to lock (you will hear a click and feel the resistance of a spring)

Mounting the sensor on the sled

A metal frame is mounted to the front of the sled, featuring threaded holes in each of its four corners. One sensor is mounted to a plastic plate with matching holes and is to be mounted with the sensor facing inward, and the wire going out to the right if you are looking at the front of the sled. The second sensor is attached to a magnet and should be placed in the back of the sled near the weights, above the brain box, and centered. The brain should be in the back, near the weights, with a cushion beneath it to restrict its movement and reduce the vibration it is subjected to. Tape down the box as necessary and ensure the wire for the front sensor goes over the top of the sled, and not around the side, to ensure it doesn't get caught or broken while the sled is moving.

How to clean sensor

1. Unscrew the 4 allen key screws on the front panel of the sled to remove the plastic cover with sensor and place to the side
2. Clean off the concrete that is stuck to the mount by using a chisel

3. Using a smaller chisel clean out the hollow bars on either side of the mount
4. Brush off concrete from the sensor on the plastic cover and the sensor on the magnet if necessary
5. Replace cover with the sensor facing inward and the wire going out to the right if facing the front of the vibrating sled